

STUDENT PLACEMENT PREDICTION SYSTEM

A MACHINE LEARNING BASED PLACEMENT PREDICTION APPLICATION

PRESENTATION

Submitted By:

Ritika Gauttam

Internship Organization:

Upflairs Pvt. Ltd.

Internship Domain:

Data Science with Machine Learning and AI

INTRODUCTION

What is Student Placement Prediction?

Student Placement Prediction is a machine learning application that predicts whether a student is likely to get placed based on academic performance, technical skills, internships, projects, and other relevant factors.

Objective

- Predict placement status of students.
- Identify factors affecting placement.
- Assist students in understanding areas for improvement.
- Demonstrate end-to-end machine learning deployment.

PROBLEM STATEMENT

Problem

Many students are uncertain about their placement readiness.

Solution

Develop a machine learning model that analyzes student profiles and predicts placement outcomes based on various academic and skill-related attributes.

DATASET DESCRIPTION

- FeatureDescription
- Branch
- Student branch
- College Tier
- Tier-1, Tier-2, Tier-3
- CGPA
- Academic performance
- Backlogs
- Number of active backlogs
- Coding Skills
- Coding proficiency score
- DSA Score
- Data Structures & Algorithms score
- Aptitude Score
- Aptitude assessment score
- Communication Skills
- Communication rating
- ML Knowledge
- Machine Learning knowledge score
- System Design
- System design understanding
- Internships
- Number of internships
- Projects Count
- Number of projects completed
- Certifications
- Number of certifications

Target Variable

- Placement Status
 - 1 = Placed
 - 0 = Not Placed

DATA PREPROCESSING

Steps Performed

- Checked missing values.
- Removed unnecessary columns.
- Encoded categorical variables.
- Converted college tiers into numerical values.
- Prepared data for machine learning.
- Split dataset into training and testing sets.

Example Encoding

Tier	Encoded Value
Tier-3	0
Tier-2	1
Tier-1	2

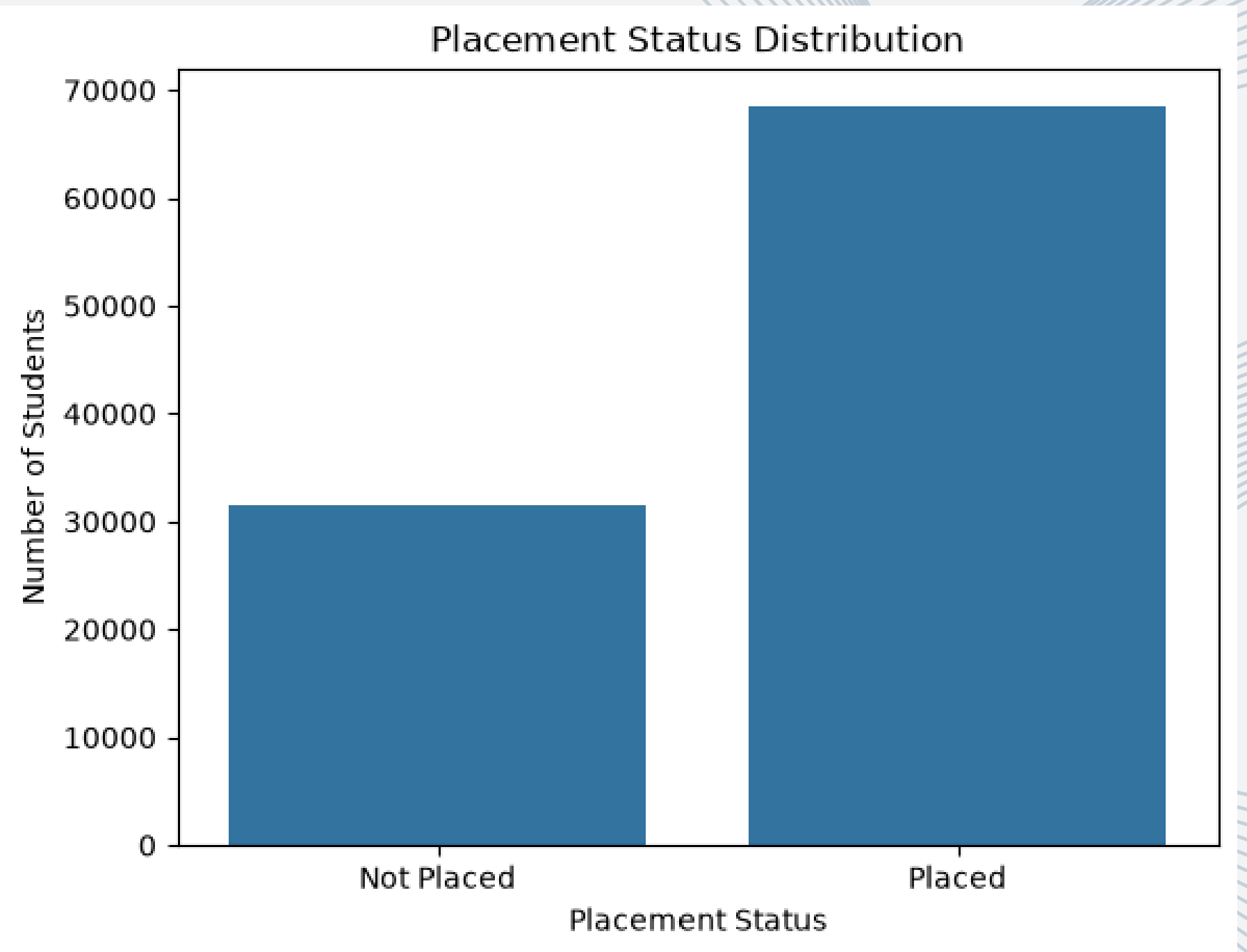
EXPLORATORY DATA ANALYSIS (EDA)

Analysis Performed

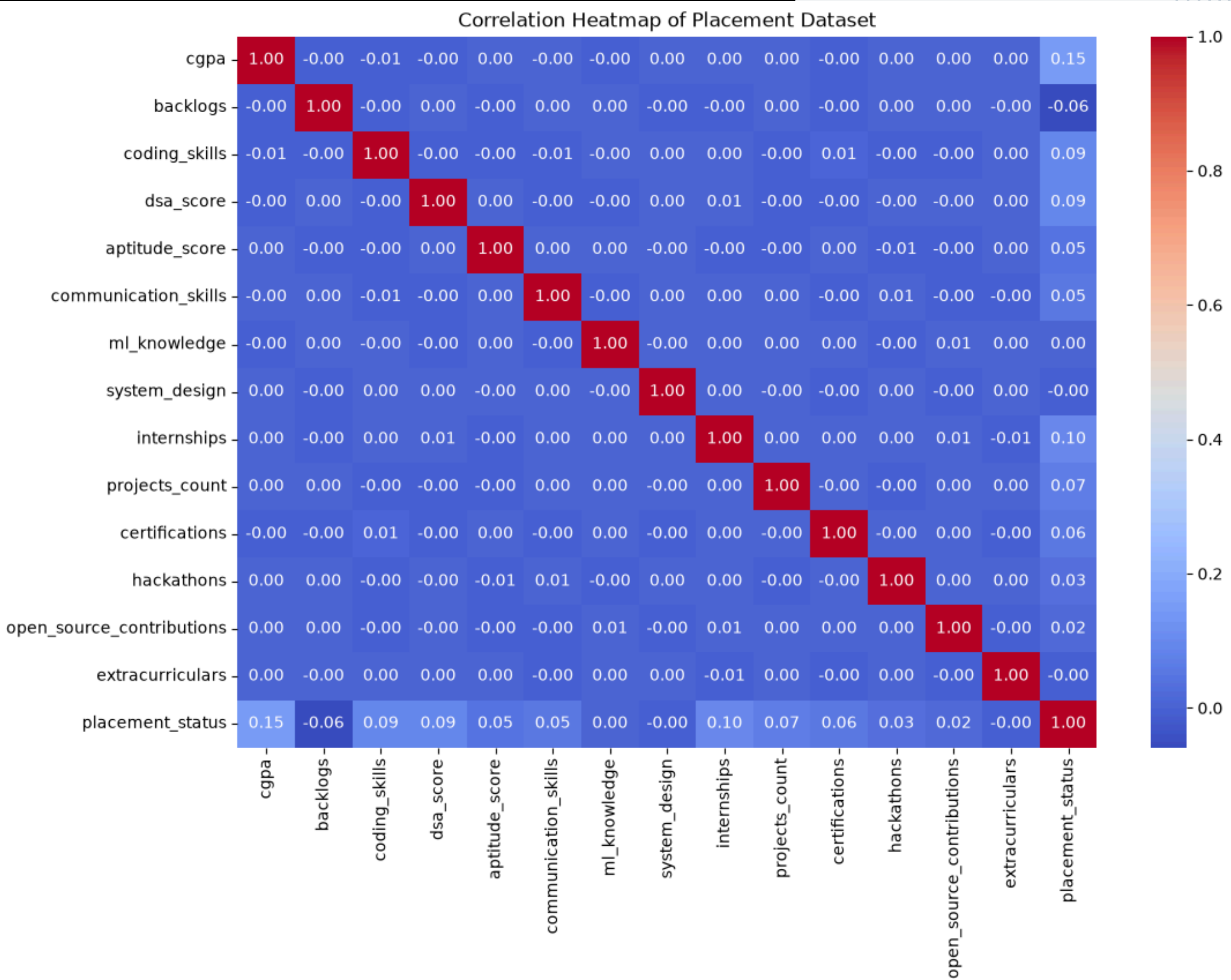
- Distribution of placement status.
- Relationship between CGPA and placement.
- Impact of internships on placement.
- Effect of coding and DSA skills.
- Correlation analysis among features.

Key Observation

Students with higher CGPA, strong coding skills, internships, and project experience showed a higher probability of placement.



EXPLORATORY DATA ANALYSIS (EDA) 06



MACHINE LEARNING MODEL

Algorithm Used

Random Forest Classifier

Why Random Forest?

- Handles complex relationships.
- Reduces overfitting through ensemble learning.
- Performs well on structured datasets.
- Provides robust predictions.

Hyperparameters

n_estimators = 200

max_depth = 12

min_samples_split = 5

min_samples_leaf = 2

MODEL TRAINING PROCESS

Workflow

Dataset



Preprocessing



Train-Test Split



Random Forest Training



Prediction



Evaluation

Split Ratio

- Training Data: 80%
- Testing Data: 20%

MODEL PERFORMANCE

RESULTS

Metric	Value
Training Accuracy	75.00%
Testing Accuracy	70.00%

Interpretation

- Model generalizes reasonably well.
- Limited overfitting compared to earlier models.
- Suitable for educational prediction purposes.

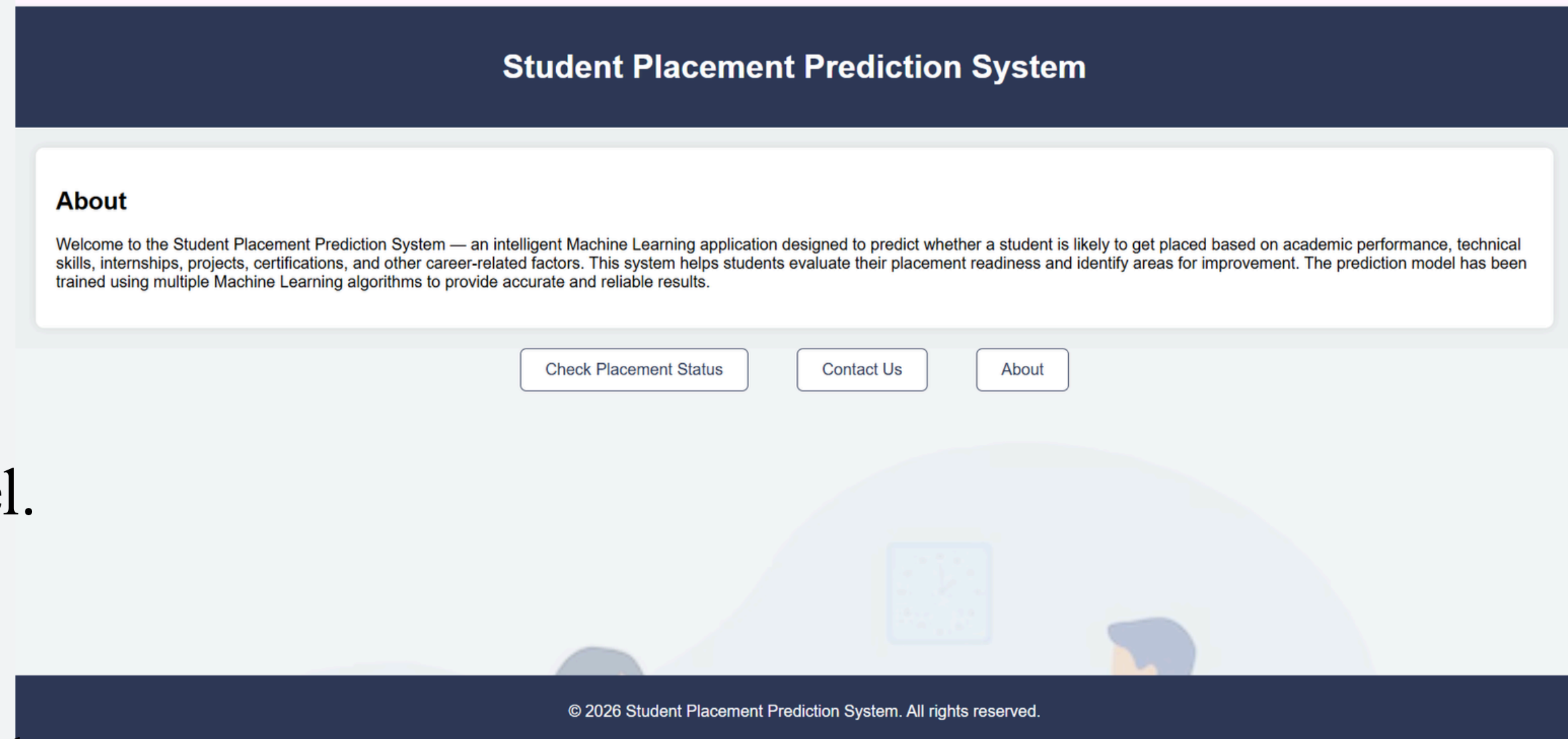
DEPLOYMENT

Technologies Used

- Flask
- HTML
- Joblib
- CSS

Deployment Process

1. Train Random Forest Model.
2. Save model using Joblib.
3. Build Flask backend.
4. Create user-friendly frontend.
5. Take student inputs.
6. Generate placement prediction.



DEPLOYMENT

Placement Predictor

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← → ↻ ⓘ 127.0.0.1:5000/contact

This is my number :-6375458220

About Student Placement Prediction System

The Student Placement Prediction System is a Machine Learning based web application designed to predict whether a student is likely to get placed based on their academic performance, technical skills, internships, projects, certifications, and other important factors.

The objective of this project is to help students understand their placement readiness and identify areas where improvement may increase their chances of securing job opportunities.

This system analyzes multiple student attributes including CGPA, coding skills, aptitude score, communication skills, internships, projects, certifications, hackathons, open-source contributions, and extracurricular activities.

Various Machine Learning algorithms such as Support Vector Machine (SVM), Decision Tree, K-Nearest Neighbors (KNN), Random Forest, and Naive Bayes were trained and evaluated. The best-performing model was selected and deployed using Flask to provide real-time placement predictions through an interactive web interface.

This project demonstrates the practical application of Data Science, Machine Learning, Data Analysis, and Web Development in solving real-world educational and career-oriented problems.

Project Features:

- Student Placement Prediction
- Machine Learning Based Decision Making
- User-Friendly Flask Web Application
- Real-Time Prediction Results
- Data-Driven Career Insights

WORKING DEMONSTRATION

User Inputs

- Branch
- College Tier
- CGPA
- Coding Skills
- DSA Score
- Internships
- Certifications
- etc.

Output

- Placed
- Not Placed

Student Placement Prediction System

Enter Student Details

Branch

Select branch

College Tier

Select college tier

CGPA

1.0 - 10.0

Backlogs

Select number of backlogs

Coding Skills

1.0 - 10.0

DSA Score

1.0 - 10.0

Aptitude Score

20.0 - 100.0

Communication Skills

Enter value between 1.0 and 10.0

ML Knowledge

0.0 - 10.0

System Design

0.0 - 10.0

Internships

Select number of internships

Projects Count

Select number of projects

Certifications

Select number of certifications

Predict Placement

Student Placement Prediction System

Enter Student Details

Branch

Computer Science Engineering (CSE)

College Tier

Tier-2

CGPA

9

Backlogs

0

Coding Skills

8

DSA Score

8

Aptitude Score

70

Communication Skills

8

ML Knowledge

6

System Design

5

Internships

2

Projects Count

4

Certifications

4

Predict Placement

Placed

CHALLENGES FACED

Challenges

- Feature selection.
- Handling categorical data.
- Avoiding overfitting.
- Aligning frontend inputs with model features.
- Flask deployment debugging.

Solutions

- Feature engineering.
- Hyperparameter tuning.
- Proper validation.
- Input validation in HTML forms.

FUTURE SCOPE

Possible Enhancements

- Predict salary package.
- Add resume analysis.
- Include interview performance score.
- Deploy on cloud platforms.
- Develop a complete career recommendation system.

The Student Placement Prediction System successfully predicts placement outcomes using machine learning techniques. The project demonstrates the complete machine learning lifecycle, including data preprocessing, model training, evaluation, and web deployment.

Key Learning

- Data Preprocessing
- EDA
- Feature Engineering
- Machine Learning
- Flask Deployment
- Frontend Integration

THANK YOU